Part A:

```
#include <iostream>
using namespace std;
double calculateFine(int daysLate)
{
    return daysLate * 5;
}
int main()
{
    int daysLate;
    cout << "Enter number of days late: ";
    cin >> daysLate;
    cout << "Fine: " << calculateFine(daysLate);
    return 0;
}
```

Category	**Details**
Purpose of the program	Calculates and displays the total late fine based on the number of overdue days entered by the user.
Input accepted	An integer (int) representing daysLate (the number of days overdue).
Assumption made	Assumes a flat rate of 5 units of currency per late day and assumes the user will only enter non-negative integer values.
Invalid input	Non-numeric characters (e.g., "five", "abc"), floating-point numbers (e.g., 3.5), or negative values (e.g., -4).
Edge case	Entering 0 days late (which yields a fine of 0), or entering extremely large integers that could cause integer overflow.
Suggested improvement	Add input validation to ensure the entry is a non-negative integer and reject invalid inputs (e.g., using cin.fail() or an if (daysLate < 0) check).